

Yinghao (Peter) Guan

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Summary

A Computer Science student with hands-on experience in data analysis, AI applications, and full-stack development. Delivered 8+ academic and independent projects, including an AI-powered language learning game and a behavioural economics study using PLS-SEM. Adept in Python, Java, and C#, eager to contribute analytical and problem-solving skills to a software or data-focused internship.

Education

Santa Monica College , Computer Science	Sept 2025 – Present
University of Edinburgh , BEng in Computer Science	Sept 2022 – July 2024
Relevant Coursework: Software Engineering, Data Structure, Reasoning and Agents	

Projects and Activities

Custom Markup Language Editor & Converter	Oct 2025
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- Developed a desktop application using **JavaFX** and **ANTLR4** to parse a custom Markdown syntax into an Abstract Syntax Tree (AST), demonstrating core compiler principles.
- Implemented a visitor pattern to transform the AST into a JSON-formatted Pandoc AST, which was then piped to the **Pandoc CLI** for conversion to HTML and LaTeX, showcasing systems integration.
- Followed **MVVM architecture** to separate view from business logic, improving the extensibility and testability of the application.

UCLA HCI Research Collaboration	Jul 2025 – Aug 2025
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- Collaborated with UCLA HCI faculty on an exploratory research project, delivering presentations and preliminary findings.
- Preparing a peer-reviewed paper for future submission.

Independent Research: Behavioural Modelling of Excessive Trading in Tech Stocks	Jan 2025 – Apr 2025
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- Collected and integrated US tech sector stock data (META, GOOGL, MSFT) with sentiment indicators using Python (pandas, yfinance).
- Modelled the impact of DeepSeek R1 LLM launch on trading behaviour via PLS-SEM, revealing significant FoMO and overconfidence effects.
- Produced visualisations in Matplotlib to illustrate behavioural bias patterns across event windows.

Data Analysis: Performance Trends in Ultra Marathon Runners	Sep 2023 – Dec 2023
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- Processed and analysed a Kaggle dataset with 200k+ race records using Python (pandas, seaborn, sklearn, scipy).
- Applied K-means clustering to segment runners by performance and demographics.
- Visualised trends across race categories, highlighting club-affiliated and age-related patterns.

Additional Information

Computer Skills: Python, Java, R, C, C#, Haskell, Unity2D, LaTeX, Git

Languages: Chinese, English